26900563 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Skill Enhancement Course (SEC) - MG5SECZGY301 - AQUARIUM FABRICATION AND SETTING

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Which sequence correctly describes nitrification in an aquarium? [U] / [CO2]

   a). Nitrate---> Nitrite---> Ammonia

   b). Nitrite----> Nitrate-----> Ammonia

   c). Ammonia----> Nitrite-----> Nitrate

   d). Ammonia----> Nitrate-----> Nitrite

2. Which of the following is correctly classified as a furniture-integrated aquarium? [K] / [CO1]

   a). Wall niche aquarium

   b). Coffee table aquarium

   c). Room divider aquarium

   d). Floor-integrated tank

3. Infusoria are mainly used as [K] / [CO3]

   a).Live food organisms

   b).Aquarium decorations

   c).Water disinfectants

   d).Fish medicines

4. Dropsy disease is caused by [U] / [CO4]

a). Aeromonas b). Saprolegnia

c). Pseudomonas d). Argulus

5. Air stones help to improve ----------. [U] / [CO2]

a). Removal of dissolved metabolic wastes

b). Maintenance of constant water temperature

c). Lighting

d). Aeration by producing tiny air bubbles

6. Aquascaping style dominated by carefully arranged stones is [U] / [CO3]

a). Iwagumi style b). Jungle style

c). Dutch style style d). Paludarium style

7. Where is HOB filter primarily mounted? [K] / [CO2]

a). Under the substrate b). Inside the tank

c). On the back wall of the aquarium d). On the water surface

8. The substrate commonly used in marine aquariums because it helps maintain alkalinity and stabilize pH is [U] / [CO3]

a). Aragonite sand b). clay soil

c). Garden soil d). Peat moss

9. What is the generally recommended duration of lighting for an aquarium per day? [U] / [CO4]

a). 2–4 hours b). 5–6 hours

c). 8–10 hours d). 12–14 hours

10. Zeolite is particularly useful for removing ----------. [U] / [CO2]

a). Carbon dioxide b). Nitrogen gas

c). Ammonia d). Organic waste

11. Which sequence correctly matches the purpose with the appropriate speciality aquarium? [An] / [CO1]

a). Scientific study — Public display aquarium	b). Public viewing — Experimental aquarium	c). Scientific research — Experimental or research aquarium	d). Maintaining jellyfish — Planted aquarium

12. What is the main purpose of a battery backup in an aquarium? [K] / [CO2]

a). Maintain constant water temperature

b). Provide power to aquarium equipments during power failures

c). Remove metabolic wastes

d). Dispense food at regular intervals

[1x12 = 12]

Part B

Short Answer Questions

Answer any **6** questions

Each question carries **3** marks

13. What is meant by feeding rate? [K] / [CO3]

14. Why are moderately sized or large aquariums easier to maintain than very small aquariums? [An] / [CO3]

15. Write an account on feeding rings and their advantages. [K] / [CO2]

16. State the major characteristic feature of a planted aquarium. [K] / [CO1]

17. Mention the role of aquarium plants in nitrogen cycle. [K] / [CO3]

18. Name any six common aquatic plants used in aquarium industry. [U] / [CO3]

19. What is the importance of maintaining adequate oxygen during fish transportation? [U] / [CO4]

20. List the different types of aquarium heaters and mention the key features that distinguishes them from each other. [U] / [CO2]

[3x6 = 18]

Part C

Short Essay Questions

Answer any **4** questions

Each question carries **5** marks

21. Describe the different shapes of aquariums commonly used in aquarium fabrication. [U] / [CO1]

22. Explain the significance of regular aquarium maintenance and the tools essential for maintaining a healthy aquatic environment. [U] / [CO2]

23. Discuss the different types of dry and non-dry feeds used in aquarium fish culture. [U] / [CO3]

24. Explain the concept of fish compatibility and discuss the factors that should be considered when selecting fishes for an aquarium. [U] / [CO4]

25. Describe different types of aquarium rocks and comment on their significance. [K] / [CO2]

26. An aquarist wishes to maintain fish together with live rock in a saltwater aquarium. Identify the most suitable type of aquarium and justify your answer. Explain its main characteristics. [A] / [CO1]

[5x4 = 20]